

EDUCATION

- **Massachusetts Institute of Technology** Cambridge, MA
Ph.D. Candidate in Physics Aug. 2024 - Present
- **Harvard College** Cambridge, MA
A.B. in Physics and Astrophysics Aug. 2019 - May 2024
Summa Cum Laude; GPA: 4.00/4.00
Senior Thesis: Inverse Design of Complex Fluids with Fully-Differentiable Lagrangian Particle Dynamics
- **Lynbrook Senior High School** Lynbrook, NY
Valedictorian; GPA: 105.2/100 Sep. 2015 - June 2019

GRADUATE FELLOWSHIPS

- **Hertz Fellowship**
- **National Science Foundation Graduate Research Fellowship**
- **Ida M. Green Fellowship at MIT**

AWARDS

- **Sophia Freund Prize (2024)**
- **[Thomas T. Hoopes Prize](#) (2024)**
- **[Bowdoin Prize in the Natural Sciences](#) (2023)**
- **Phi Beta Kappa (Junior 24)**
- **Sosland Prize in Expository Writing (2020)**
- **Sally Ride Internship Award at NASA (2021)**
- **Undergraduate Research Fellowships:** Herchel-Smith Fellowship (2020 & 2021) and Program for Research in Science and Engineering (2020 & 2021)

INTERNSHIP EXPERIENCE

- **Google Research** Cambridge, MA
Science AI Team June 2026 - Present
- **NASA Ames Research Center** Remote
Data Science Group (ExoMiner Team) Jan. 2021 - May 2022
- **NASA Langley Research Center** Remote
Computational Fluid Dynamics Group Aug. 2020 - Dec. 2020
- **Citadel** New York, NY
Global Fixed Income (Trading) Jun. 2022 - Aug. 2022

PUBLICATIONS

- **[HARDMath: A Benchmark Dataset for Challenging Problems in Applied Mathematics](#)**; Jingxuan Fan*, Sarah Martinson*, Erik Wang*, **Kaylie Hausknecht***, Jonah Brenner, Danxian Liu Nianli Peng, Corey Wang, Michael Brenner (2024). ICLR 2025.
- **[Control of Flow Behavior in Complex Fluids Using Automatic Differentiation](#)**; Mohammed Alhashim, **Kaylie Hausknecht***, Michael Brenner (2024). Proceedings of the National Academy of Sciences of the United States of America, 122(8).
- **[ExoMiner: A Highly Accurate and Explainable Deep Learning Classifier that Validates 301 New Exoplanets](#)**; Hamed Valizadegan, Miguel Martinho, Laurent S. Wilkens, Jon M. Jenkins, Jeffrey Smith, Douglas A. Caldwell, Joseph D. Twicken, Pedro C. Gerum, Nikash Walia, **Kaylie Hausknecht***, Noa Y. Lubin, Stephen T. Bryson, Nikunj C. Oza (2022). *The Astrophysical Journal*, 926(2), 120.

*Equal contribution

CONFERENCE PRESENTATIONS

- [HARDMath: A Benchmark Dataset for Challenging Problems in Applied Mathematics](#)
ICLR, Singapore, April 2025.
- [HARDMath: A Benchmark Dataset for Challenging Problems in Applied Mathematics](#)
NeurIPS Math AI Workshop.
- **Inverse Design of Complex Fluids with Fully Differentiable Particle Dynamics** Institute for Artificial Intelligence and Fundamental Interactions Summer Workshop, Cambridge, MA, August 2024.
- [Optimizing Mixing in Porous Media via Automatic Differentiation](#)
American Physical Society Division of Fluid Dynamics Annual Meeting, Washington DC, November 2023.
- [Optimizing Chaotic Mixing with Automatic Differentiation](#)
AI Institute in Dynamic Systems, Seattle, WA, July 2023.
- [Machine Learning Correlates Charge Density Wave with the Local Gap in Cuprate Superconductors](#)
American Physical Society March Meeting, Virtual, March 2021.
- [Machine Learning Correlates Charge Density Wave with the Local Gap in Cuprate Superconductors](#)
American Physical Society Conference for Undergraduate Women in Physics, Virtual, January 2021.
- [Local correlations in disordered materials with neural networks](#)
Machine Learning Quantum Matter Data Simons Foundation, January 2020
- [Machine Learning Correlates CDW Properties with Local Gap in Cuprates](#)
American Physical Society March Meeting, Boston, MA, March 2019.
- [Identifying Inhibitors of T-Cell Activation in Microgravity: Proposal for the International Space Station](#)
International Space Station R&D Conference, San Francisco, CA, July 2018.

TEACHING ASSISTANT

- **MIT**
 - Statistical Field Theory (8.334) Spring 2026
- **Harvard**
 - Physical Mathematics (Applied Math 201) Fall 2023
 - Introductory Mechanics and Relativity (Physics 15a) Fall 2021
 - Linear Algebra and Real Analysis (Math 23a) Summer & Fall 2020
 - Theory of Calculus (Math 370) Aug. 2020